GEN 2 TABLES AND CODES

GEN 2.1 MEASURING SYSTEM, AIRCRAFT MARKINGS, HOLIDAYS

1 Units of measurement

The units of measurement used in connection with all aspects of civil aviation air and ground operations in the Republic of South Africa are in accordance with the Civil Aviation Regulations, 2011, Part 1, sub-part 2 Units of measurement.

ampere (A) is the constant electric current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 metre apart in vacuum, would produce between these conductors a force equal to 2 x newton per metre of length;

becquerel (Bq) is the activity of a radionuclide having one spontaneous nuclear transition per second;

candela (cd) is the luminous intensity, in the perpendicular direction, of a surface of 1/600 000 square metre of black body at the temperature of freezing platinum under a pressure of 101 325 newtons per square metre;

celsius temperature (tºc) is equal to the difference tºc =T - To between two thermodynamic temperatures T and To where To equals 273.15 kelvin;

coulomb (C) is the quantity of electricity transported in 1 second by a current of 1 ampere;

degree celsius (ºC) is the special name for the unit kelvin for use in stating values of Celsius temperature;

farad (F) is the capacitance of a capacitor between the plates of which there appears a difference of potential of 1 volt when it is charged by a quantity of electricity equal to 1 coulomb;

foot (ft) is the length equal to 0.3048 metres exactly;

gray (Gy) is the energy imparted by ionising radiation to a mass of matter corresponding to 1 Joule per kilogram;

henry (H) is the Inductance of a closed circuit in which an electromotive force of 1 volt is produced when the electric current in the circuit varies uniformly at a rate of 1 ampere per second;

hertz (Hz) is the frequency of a periodic phenomenon of which the period is 1 second;

joule (J) is the work done when the point of application of a force of 1 newton is displaced a distance of 1 metre in the direction of the force;

kelvin (K) is the unit of thermodynamic temperature which is the fraction 1/273.16 of the thermodynamic temperature of the triple point of water;

kilogram (kg) is the unit of mass equal to the mass of the international prototype of the kilogram;

knot (kt) is the speed equal to 1 nautical mile per hour;

litre (L) is a unit of volume restricted to the measurement of liquids and gases which is equal to 1 cubic decimeter;

lumen (lm) is the luminous flux emitted in a solid angle of 1 steradian by a point source having a uniform intensity of 1 candela;

lux (lx) is the illuminance produced by a luminous flux of 1 lumen uniformly distributed over a surface of 1 square metre;

metre (m) is the distance travelled by light in a vacuum during 1/299792458 of a second;

mole (mol) is the amount of substance of a system which contains as many elementary entitles as there are atoms in 0.012 kilogram of carbon-12;

Note.-When the mole is used, the elementary entities must be specified and may be atoms, molecules, ions, electrons, other particles or specified groups of such particles.

nautical mile (NM) is the length equal to 1 852 metres exactly;

newton (N) is the force which when applied to a body having a mass of 1 kilogram gives it an acceleration of 1 metre per second squared;

ohm (Ω) is the electric resistance between two points of a conductor when a constant difference of potential of 1 volt, applied between these two points, produces In this conductor a current of 1 ampere, this conductor not being the source of any electromotive force;

pascal (Pa) is the pressure or stress of 1 newton per square metre;

radian (rad) is the plane angle between two radii of a circle which cut off on the circumference an arc equal In length to the radius;

second (s) is the duration of 9 192 631 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium-133 atom;

siemens (S) is the electric conductance of a conductor in which a current of 1 ampere is produced by an electric potential difference of 1 volt;

sievert (Sv) is the unit of radiation dose equivalent corresponding to 1 joule per kilogram;

steradian (sr) is the solid angle which, having its vertex in the centre of a sphere, cuts off an area of the surface of the sphere equal to that of a square with sides of length equal to the radius of the sphere;

tesla (T) is the magnetic flux density given by a magnetic flux of 1 weber per square metre;

tonne (t) is the mass equal to 1 000 kilograms;

volt (V) is the unit of electric potential difference and electromotive force which is the difference of electric potential between two points of a conductor carrying a constant current of 1 ampere, when the power dissipated between these points is equal to 1 watt;

watt (W) is the power which gives rise to the production of energy at the rate of 1 Joule per second; and

weber (Wb) the magnetic flux which, linking a circuit of one turn produces in it an electromotive force of 1 volt as it is reduced to zero at a uniform rate in 1 second.

2 Temporal reference system

Universal Coordinated Time (UTC) is used by air navigation services and in documents published by the aeronautical information service. Reporting of time is expressed to the nearest minute, e.g. 12: 40: 35 is reported as 1241. Local time is UTC + 2 Hours.
The Gregorian calendar shall be used as the temporal reference system for air navigation.

3 Geodetic reference datum

3.1 Name/designation of datum.
The geographical coordinates indicating Latitude and Longitude are expressed in terms of the World Geodetic System of 1984 (WGS-84) geodetic reference datum.

3.2 Area of application.
The area of application for the published geographical coordinates coincides with the area of responsibility of the Aeronautical Information Service, i.e. the entire territory of the RSA as well as the airspace over the high seas encompassed by the Johannesburg Oceanic FIR in accordance with the regional air navigation agreement.

3.3 Use of an asterisk to identify published geographical coordinates.
The application of WGS-84 will be achieved either by surveyor mathematical conversion of coordinates. Where the information has been transformed mathematically into WGS-84 coordinates or where the accuracy of original field work does not meet the ICAO Standards and Recommended Practices they are published accompanied by an asterix (*) indicating that the information is of low integrity.

3.4 In addition to elevation referenced to the MSL (geoid), for the specific surveyed ground positions, geoid undulation (referenced to the WGS-84 ellipsoid) for those positions specified in ICAO Doc 10066 Appendix 2 shall also be published.

4 Aircraft nationality and registration marks

The nationality marks for aircraft registered in the Republic of South Africa are the letters ZS, ZT and ZU. The nationality marks are followed by a hyphen and a registration mark consisting of 3 letters, e.g. ZS-ABC, ZT-ABC, ZU-ABC.

5 Public holidays for the year 2026.

Date	Day	Name
01 January	Thursday	New Year's Day
21 March	Saturday	Human Rights Day
03 April	Friday	Good Friday
06 April	Monday	Family Day
27 April	Monday	Freedom Day
01 May	Friday	Workers Day
16 June	Tuesday	Youth Day
09 August	Sunday	National Women's Day
10 August	Monday	Public Holiday
24 September	Thursday	Heritage Day
16 December	Wednesday	Day of Reconciliation
25 December	Friday	Christmas Day
26 December	Saturday	Day of Goodwill